

Practical - 1

Network details

- **Router0** and **Router1** are connected by a **WAN / point-to-point link** (network **10.0.0.0**).
- Each router connects to one or more **LANs** via switches.
- Each LAN has **end devices** (PCs and laptops).
- **Router0 WAN IP:** **10.0.0.1**
- **Router1 WAN IP:** **10.0.0.2**

LAN 1 (Router0)

- Network: **192.168.1.0/24**
- Router0 LAN IP (gateway): **192.168.1.2**
- Devices:
 - Laptop0: **192.168.1.10**
 - PC0: **192.168.1.11**
- Switch0 connects devices to Router0

All devices use **192.168.1.2** as default gateway

LAN 2 (Router1)

- Network: **192.168.2.0/24**
- Router1 LAN IP (gateway): **192.168.2.2**
- Devices:
 - Laptop1: **192.168.2.10**
 - PC1: **192.168.2.11**
- Switch1 connects devices to Router1

Default gateway: **192.168.2.2**

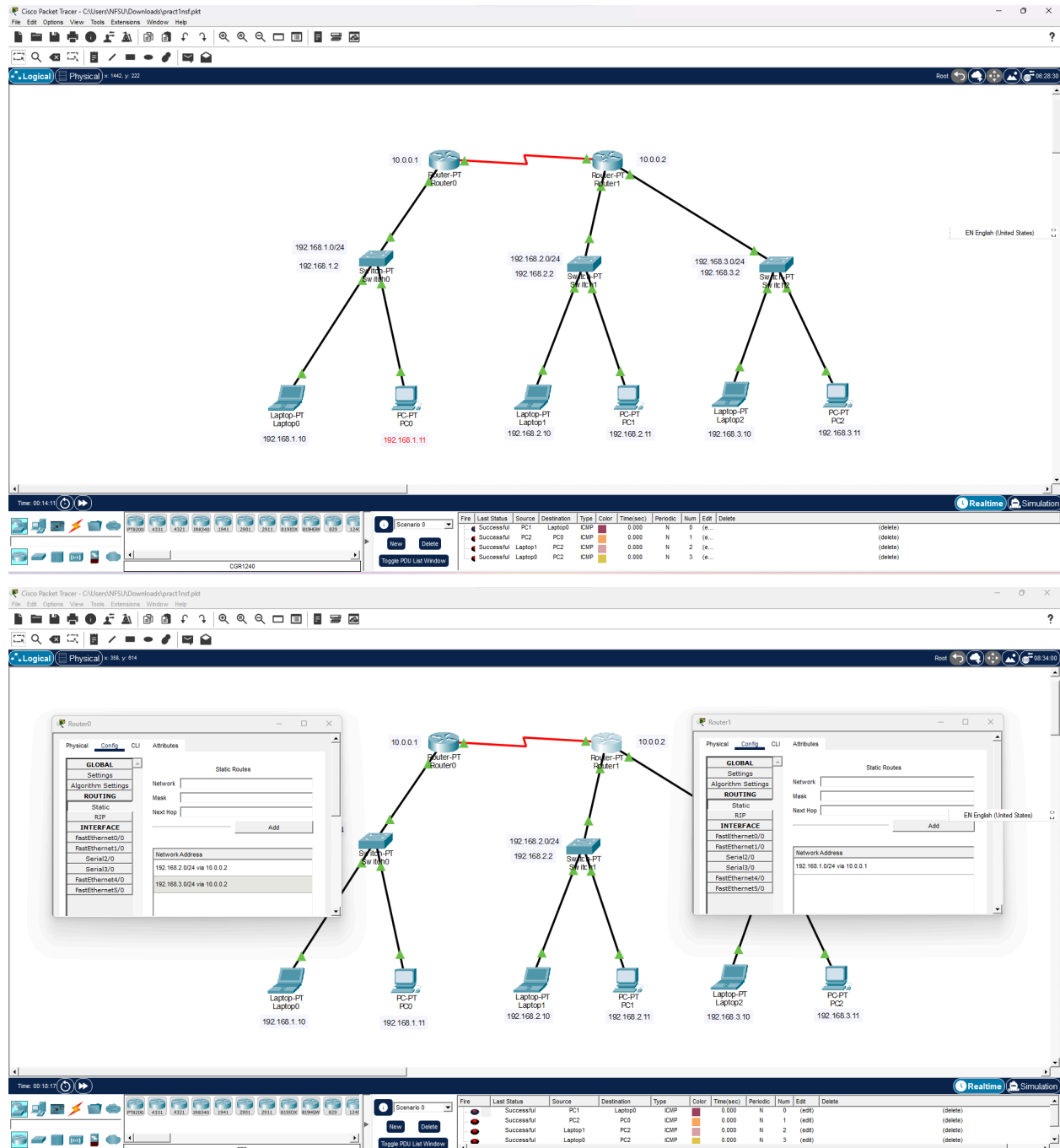
LAN 3 (Router1)

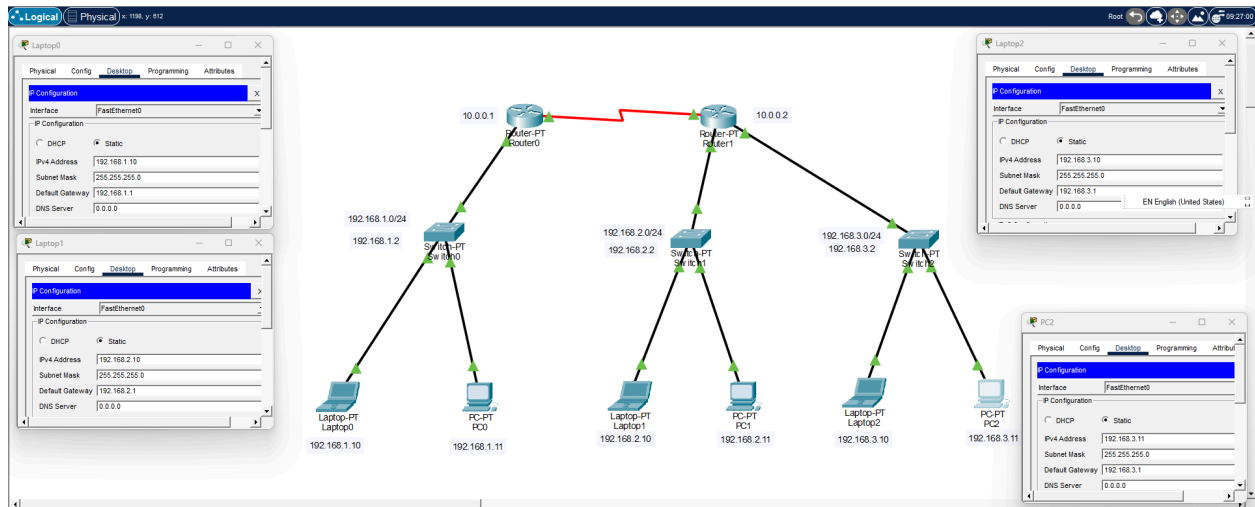
- Network: **192.168.3.0/24**
- Router1 LAN IP (gateway): **192.168.3.2**
- Devices:
 - Laptop2: **192.168.3.10**
 - PC2: **192.168.3.11**
- Switch2 connects devices to Router1

Default gateway: **192.168.3.2**

A) Static routing

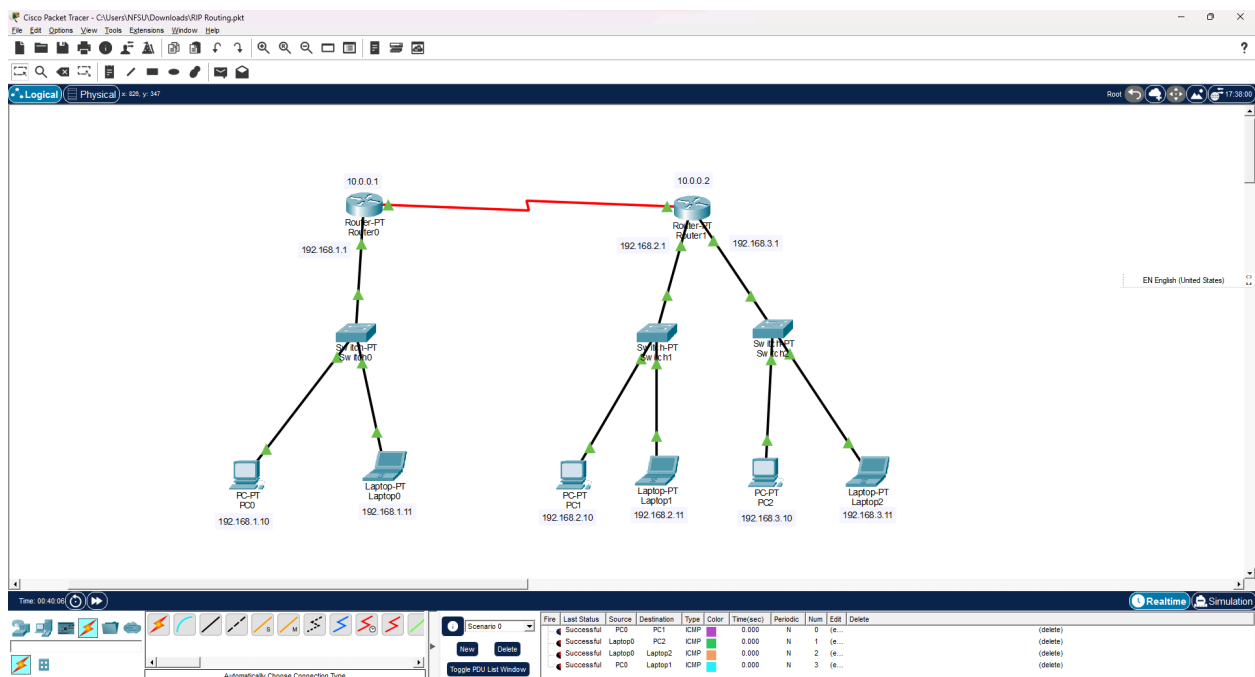
Routes are manually configured by the administrator; simple and reliable but not automatic if the network changes.

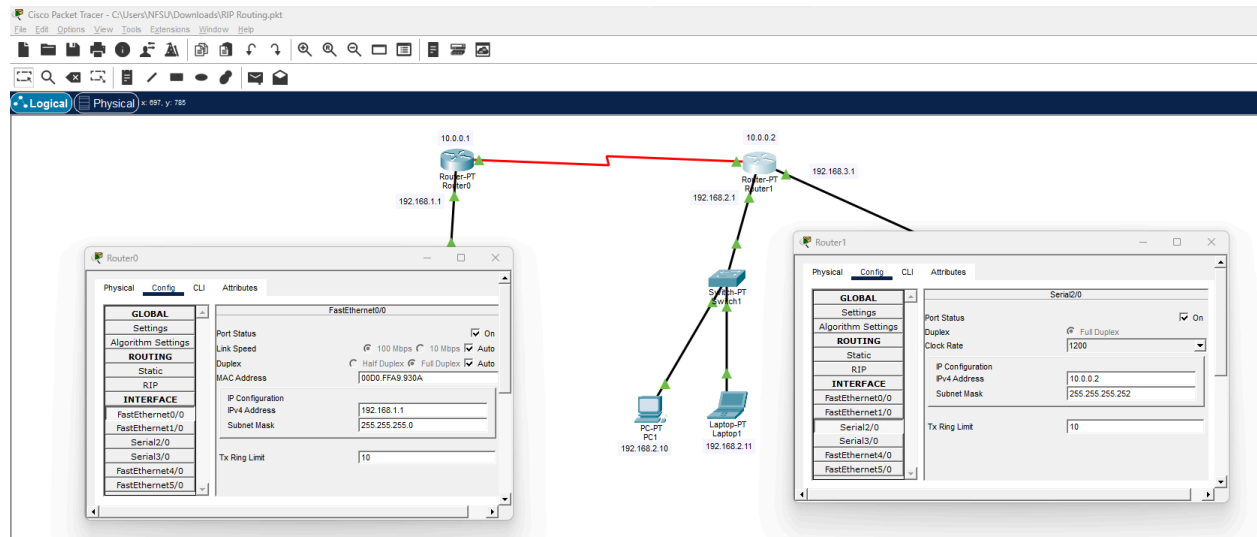
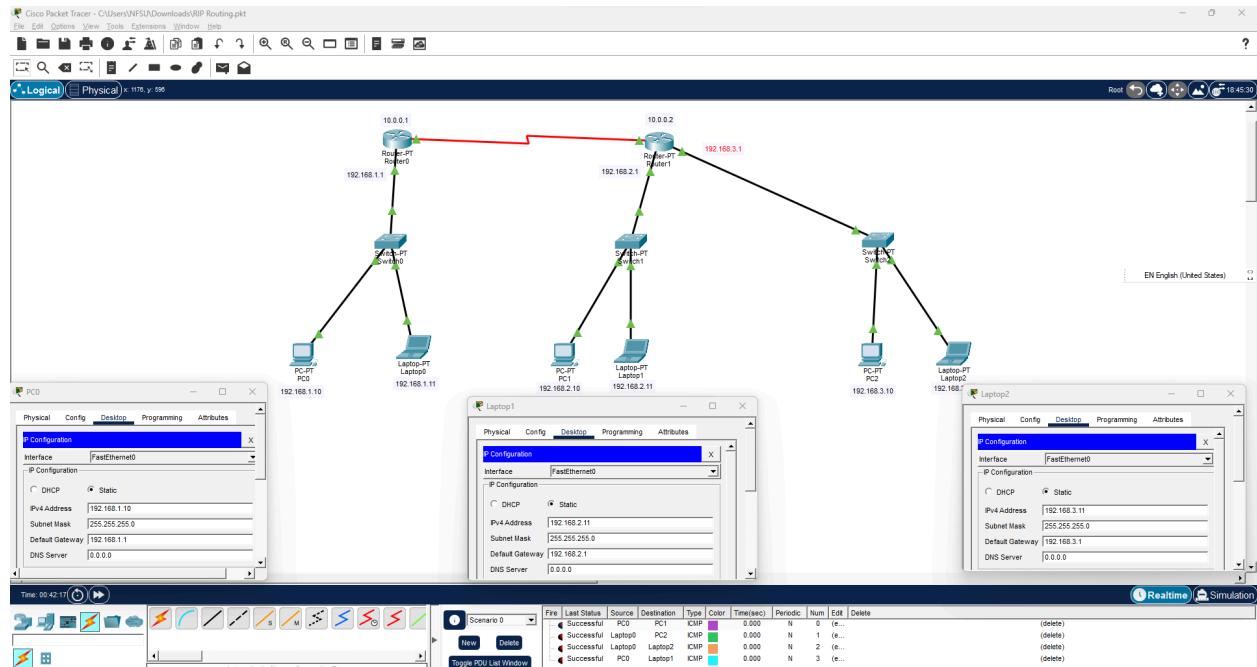


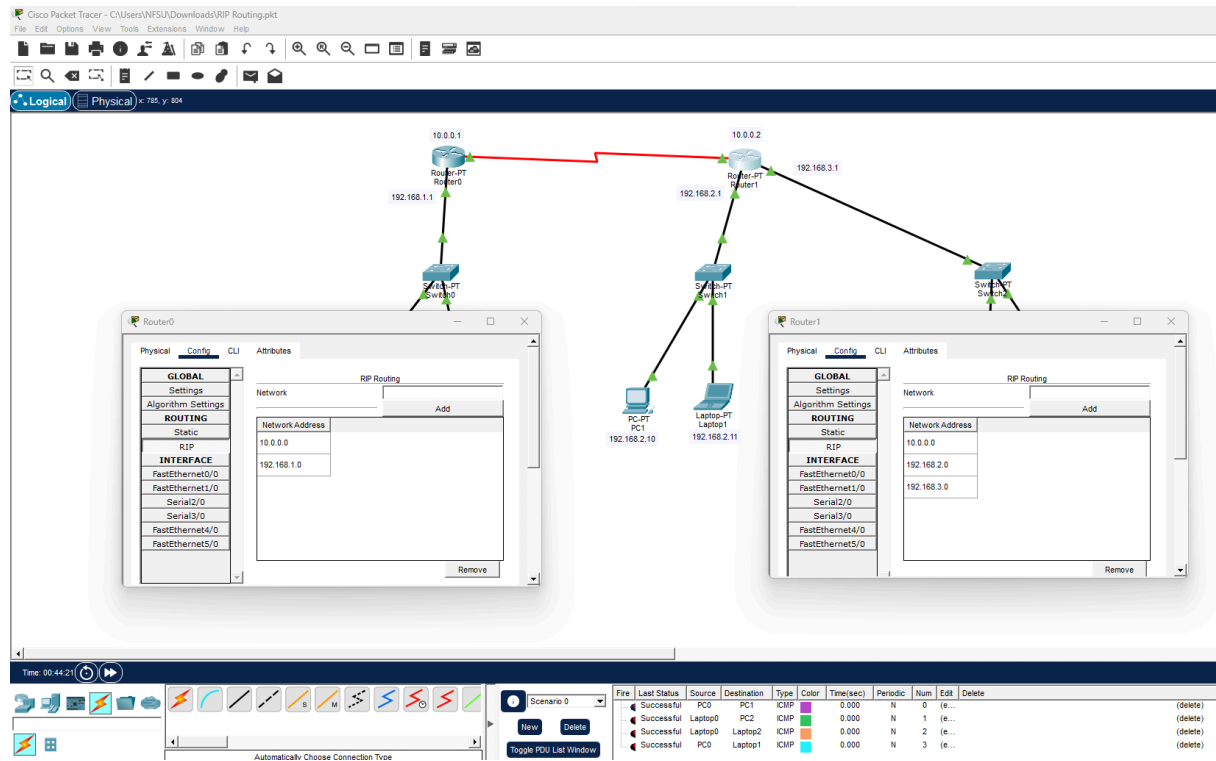


B) RIP

A dynamic routing protocol where routers automatically share routes using hop count; easy to use but slow and limited to 15 hops.





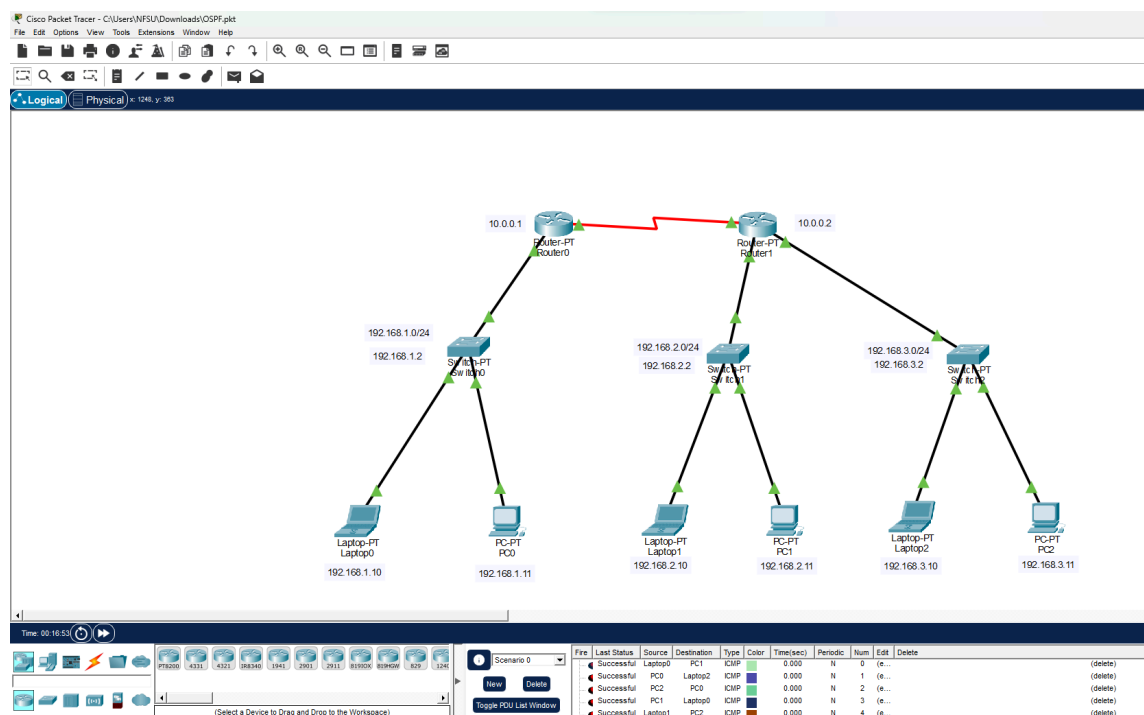


C) OSPF

OSPF is a link-state routing protocol designed for larger, more complex networks. Instead of hop counts, OSPF uses the **Shortest Path First (SPF) algorithm** (Dijkstra's algorithm) to calculate the most efficient path based on cost, which is typically tied to bandwidth.

Key Components in Your Document:

- **Router ID:** A unique identifier for the router within the OSPF network (e.g., **router-id 1.1.1.1**).
- **Areas:** OSPF uses a hierarchical structure to minimize routing overhead. "Area 0" is the backbone area that connects all other areas.
- **Wildcard Masks:** OSPF uses wildcard masks (the inverse of a subnet mask, like **0.0.0.255**) to define which IP addresses on an interface are included in the routing process.



Router0

```

Router0
Physical Config CLI Attributes
IOS Command Line Interface
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
%SYS-5-CONFIG_I: Configured from console by console

Router(config-if)#exit
Router(config)#interface FastEthernet1/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 192,168.1.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Router#
  
```

enable

configure terminal

router ospf 1

router-id 1.1.1.1

network 192.168.1.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

end

write memory

Router1

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#router-id 2.2.2.2
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
write
01:31:08: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Serial2/0 from LOADING to FULL, Loading Done
memory
Building configuration...
[OK]
Router#
```

enable

configure terminal

router ospf 1

router-id 2.2.2.2

network 192.168.2.0 0.0.0.255 area 0

network 192.168.3.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

end

write memory

Verify the protocol: *show ip protocols*

Router#show ip protocols

Routing Protocol is "ospf 1"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 1.1.1.1

Number of areas in this router is 1. 1 normal 0 stub 0 nssa

Maximum path: 4

Routing for Networks:

192.168.1.0 0.0.0.255 area 0

10.0.0.0 0.0.0.3 area 0

Routing Information Sources:

Gateway	Distance	Last Update
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1.1.1.1	110	00:02:49
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2.2.2.2	110	00:02:49
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Distance: (default is 110)

Router#